Table 2 | Signal-processing workloads and frozen evaluation cohorts

Source	Task	Computed representation	Fitting/development	Frozen evaluation
Hydraulic	Four valve-condition classes	14 channels; 28 blocks; 238 features	907 / 223 cycles	435 previously held-out cycles, grouped by operating-segment proxies
Smartphone HAR	Six activity classes	9 channels; 18 blocks; 153 features	17 / 4 official training subjects per system	2,947 windows from nine official test subjects

HAR uses the official subject split. Hydraulic grouping uses operating-segment proxies, not documented independent runs.